
Outlier Migration in Long Decode Traces: Measurement and a Failed Static Protection Intervention

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Abstract

This draft reports OutlierMigrate, a preregistered measurement study of whether high-magnitude activation channels remain stationary during long decode in the measured hybrid language models. Phase 0 passes on Granite-4.0-H-Tiny: the fraction of top-1% position-100 channels whose rank changes by more than two channel positions by decode position 10000 is 0.8178385416666667 with bootstrap 95% CI [0.797265625, 0.8368489583333334]. Phase 1 replicates the dynamic-outlier direction at larger scale on Granite-4.0-H-Small: migration fraction 0.843165650406504 with bootstrap 95% CI [0.8334349593495936, 0.8511432926829268]. A partial Phase 2 cross-family check on nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16 also passes: migration fraction 0.820809591642925 with bootstrap 95% CI [0.7865325261158594, 0.8544931149097815]. A stricter decomposition shows that strict set-leaving, not only within-set shuffling, accounts for 0.566234756098 of Phase 1 and 0.533713200380 of partial Phase 2 channels. We then test a concrete static intervention: protect the union of top-1% channels across decode positions {100, 1000, 5000, 10000} under simple symmetric INT4 weights with FP16 activations. This Phase 3 intervention kills: median recovery is 0.000000000000 with bootstrap 95% CI [0.000000000000, 0.711143244199], below the preregistered 0.20 kill floor. Ten of 24 traces also have no recoverable static gap, exceeding the preregistered 25% no-gap kill condition. The paper therefore supports robust characterization plus a negative intervention result, not a validated positive systems method.

1 Introduction

OutlierMigrate asks whether the channels carrying high-magnitude decode-time activations are fixed or whether they move during long reasoning traces. Static outliers would support static channel-protection recipes. Dynamic outliers could motivate migration-aware protection or routing methods, but only if the migration signal translates into a useful intervention rather than only a descriptive statistic.

The current paper status is deliberately limited. Phase 0 is a screen on ibm-granite/granite-4.0-h-tiny with 12 deterministic AIME-2025 reasoning traces. Phase 1 scales the same dynamic path to ibm-granite/granite-4.0-h-small with 24 deterministic traces and an extended decode-position grid. A partial Phase 2 run applies the same measurement to Nemotron-3-Nano-30B-A3B-BF16 under the authorized no-upgrade window. Phase 3 then tests the simplest migration-aware static protection intervention: preserve the union of top-1% channels across the fixed decode grid while quantizing other weights with simple symmetric INT4. That intervention fails its preregistered gate. Together, the results support a characterization paper with a negative intervention result. They do not support a camera-ready final claim, a positive systems method, an end-to-end efficiency claim, or a completed Qwen3.6/Kimi Linear validation claim. No systems-method claim remains after Phase 3: there is no demonstrated latency, throughput, memory, or quality improvement.

The contribution is deliberately narrow:

- preregistered measurement of decode-time outlier rank migration on two Granite hybrid model sizes;
- strict set-membership decomposition showing that many channels leave the protected top-1% set entirely;
- partial Nemotron-3 replication that reduces, but does not eliminate, Granite-family-specific risk;
- a preregistered negative intervention result showing that simple static union protection is insufficient under the tested quantization setup.

The paper does not claim a deployable quantization recipe, a kernel/runtime system, a performance improvement, or completed architecture-general validation.

2 Related Work

Mamba quantization and dynamic outliers. OutlierMigrate is not the first work to observe dynamic outlier behavior in Mamba-style models. In vision state-space models, QMamba reports that hidden state sequences exhibit highly dynamic variations and introduces Temporal Group Quantization for hidden states (Li et al., 2025). OuroMamba makes the point even more directly for vision Mamba models: it identifies dynamic activation-outlier variation across time steps as a failure mode for static post-training quantization and uses lightweight dynamic outlier detection during inference (Ramachandran et al., 2025). These results are important prior art. They establish that dynamic outliers are already known in the vision-Mamba setting.

Language-Mamba quantization work has emphasized a different operating regime. Mamba-PTQ shows that recurrent LLMs exhibit activation outlier channels and frames outlier-aware quantization as a first step (Pierro and Abreu, 2024). Quamba proposes a static 8-bit per-tensor recipe that suppresses maximum input values to selective SSMS and uses a Hadamard transform for output activations (Chiang et al., 2024). MambaQuant uses variance-aligned rotations and reports heavy-tailed, channel-uneven distributions in Mamba-family models (Xu et al., 2025). Quamba-SE softens the treatment of activation outliers with multiple adaptive scales but still evaluates a per-value quantizer rather than a decode-time channel migration policy (Chen and Hemani, 2026). OutlierMigrate asks which regime applies to hybrid LLMs in long reasoning traces: the static or calibration-time protection regime common in language deployment recipes, or the dynamic regime already documented in vision Mamba.

Static protection in LLM deployment. Several influential LLM quantization systems protect or transform salient activation structure using calibration-time statistics or fixed transformations. SmoothQuant migrates activation difficulty into weights using offline activation statistics (Xiao et al., 2023); AWQ protects a small set of activation-salient weight channels (Lin et al., 2023); QuaRot removes hidden state outliers with rotations (Ashkboos et al., 2024); KVQuant isolates KV cache outliers per vector and uses per-channel key quantization (Hooper et al., 2024); and BlockDialect assigns block-wise number formats from a learned formatbook (Jang and Tambe, 2025). These systems are not claimed to be wrong by our current evidence. The implication is narrower: if the protected high-magnitude set changes substantially during long decode in hybrid LLMs, static protection maps may need migration-aware updates or validation on the specific recurrent/linear-attention architecture. Several of these systems already include adaptive components, such as KVQuant’s per-vector dense-and-sparse handling or BlockDialect’s online activation format selection; our current evidence motivates stress-testing those designs on hybrid long decode, not declaring them failed.

Channel-wise gated recurrence. Recent efficient-recurrence and linear-attention architectures make data-dependent channel-wise memory control a central design axis. Gated Linear Attention (GLA) formulates linear attention as recurrent matrix-valued state with data-dependent gating (Yang et al., 2023). Gated DeltaNet combines adaptive memory erasure with delta-rule updates and reports hybrids that mix Gated DeltaNet with attention or

Mamba2 layers (Yang et al., 2024). Moonshot AI’s October 2025 Kimi Linear report is a recent example in this line: its Kimi Delta Attention (KDA) is described as channel-wise gated linear attention that extends Gated DeltaNet with finer-grained gating, and the released Kimi Linear model is a layerwise hybrid of KDA and Multi-Head Latent Attention (Moonshot AI Kimi Team, 2025). The official Qwen3.6 model card likewise describes a hybrid layout that interleaves Gated DeltaNet blocks with gated attention (Qwen Team, 2026). These sources motivate OutlierMigrate as an empirical test of whether high-magnitude channels stay fixed during long decode in hybrid language models. They do not imply that Kimi Linear, Qwen3.6, GLA, or other unmeasured models exhibit the migration measured in this draft.

3 Method

Phase 0 follows the frozen preregistration at `experimental/outlier_migrate/phase0/preregister_om_phase0.md`. The model is `ibm-granite/granite-4.0-h-tiny` at HuggingFace snapshot commit `791e0d3d28c86e106c9b6e0b4cecddee0375b6124`. The prompt set contains 12 deterministically selected AIME-2025 traces, indices 0–11, with prompt SHA `sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd` from `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json`.

Phase 1 follows the frozen preregistration at `experimental/outlier_migrate/phase1/preregister_om_phase1.md`. The model is `ibm-granite/granite-4.0-h-small` at HuggingFace snapshot commit `b8c0982bab7fde4eb48110f5a069527c008fab39`. The prompt set contains 24 deterministically selected AIME-2025 traces, indices 0–23, with prompt SHA `sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e` from `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json`.

For every transformer layer, the runner records per-channel activation magnitudes. Phase 0 records decode positions {100, 500, 1000, 5000, 10000}; Phase 1 records {100, 500, 1000, 5000, 10000, 20000}. Each gate selects the top 1% channels by mean magnitude at position 100 and computes the fraction whose rank at the final preregistered position differs by more than two channel positions. The reported gate metric is the trace-bootstrap aggregate over layers with equal layer weighting.

4 Experiments

The Phase 0 result packet is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z`. The checker output is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json`, and artifact completeness is recorded in `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/artifact_check.json`. The activation artifact SHA is `sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687` from `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json`.

The Phase 1 result packet is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z`. The checker output is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json`, and artifact completeness is recorded in `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/artifact_check.json`. The activation artifact SHA is `sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1` from `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json`.

The partial Phase 2 result packet is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z. The checker output is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/checker_result.json, the metrics file is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/metrics.json, and artifact completeness is recorded in experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/artifact_check.json. The model is nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16 at snapshot commit cbd3fa9f933d55ef16a84236559f4ee2a0526848. The activation artifact SHA is sha256:bc841b938a125f05649f84577c0ddb7ab287213a683892cb81ea6ca06296206d from experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/activation_magnitude_manifest.json. This packet is explicitly scoped as partial Nemotron-3-only validation: Qwen3.6 and Kimi Linear are deferred, and full_cross_validation_complete=false.

The Phase 3 intervention packet is experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z. It follows the frozen preregistration at experimental/outlier_migrate/phase3/preregister_om_phase3_intervention.md. The checker output is experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/checker_result.json, the metrics file is experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/metrics.json, the per-trace perplexities and recoveries are in experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/per_trace_metrics.json, and artifact completeness is recorded in experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/artifact_check.json. The model is again ibm-granite/granite-4.0-h-tiny at snapshot commit 791e0d3d28c86e106c9b6e0b4cecd0375b6124. The prompt set uses the fixed 24-trace Phase 1 set with prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e. The quantization scheme is simple symmetric per-channel INT4 weights with FP16 activations and BF16 protected-channel rows; no AWQ-style or SmoothQuant-style activation-aware scaling is used. The scored window is the 64-token window ending at decode position 10000. The preregistered recovery metric is

$$1 - \frac{\text{perplexity}_{\text{regime}} - \text{perplexity}_{\text{BF16}}}{\text{perplexity}_{\text{static1\%}} - \text{perplexity}_{\text{BF16}}}.$$

The pass rule is median recovery ≥ 0.50 with bootstrap 95% CI lower bound > 0.30 . The kill rule is recovery < 0.20 or CI upper < 0.30 . The mandatory controls are static position-100 top-2% protection and magnitude-average top-1% protection over positions {100, 1000, 5000, 10000}.

The preregistered dynamic-outlier pass rule in both gates requires migration fraction at least 0.05 and bootstrap CI lower bound greater than 0.05. This is a nonstationarity screen, not a claimed practically meaningful effect-size standard. Phase 1 uses the same dynamic path because Phase 0 returned PASS_OM_PHASE0_DECODE_TIME_MIGRATION.

5 Results

The original preregistered migration fraction intentionally conflates two effects: channels can leave the top-1% set entirely, or they can remain in the set while shuffling rank by more than two positions. For static channel protection, the stricter set-membership readout is the more direct post-hoc interpretability measure, not the preregistered gate criterion. A channel is counted as strict set-leaving only if it is in the per-trace, per-layer top-1% set at decode position 100 and has final rank greater than the top-1% boundary at decode position 10000 in Phase 0 or 20000 in Phase 1. Under that definition, 0.634244791667 of Phase 0 base top-1% channels and 0.566234756098 of Phase 1 base top-1% channels leave the protected set entirely. The within-set shuffling rates are 0.175260416667 and 0.270934959350, respectively. The gate decision remains based only on the original preregistered migration metric.

The exact Phase 0 provenance values are: model snapshot commit 791e0d3d28c86e106c9b6e0b4cecd0375b6124, prompt SHA sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f

Quantity	Phase 0	Phase 1
Decision	dynamic-outlier pass	replicated-at-scale pass
Strict set-leaving fraction	0.634244791667	0.566234756098
Strict set-leaving 95% CI	[0.605208333333, 0.664192708333]	[0.550076219512, 0.581707317073]
Within-set rank-shuffling fraction	0.175260416667	0.270934959350
Within-set rank-shuffling 95% CI	[0.160156250000, 0.191015625000]	[0.260797764228, 0.280614837398]
Original preregistered migration fraction	0.8178385416666667	0.843165650406504
Original migration 95% CI	[0.797265625, 0.8368489583333334]	[0.8334349593495936, 0.8511432926829268]
Preregistered threshold	≥ 0.05 and CI lower > 0.05	≥ 0.05 and CI lower > 0.05
Artifact complete	true	true
Model	ibm-granite/granite-4.0-h-tiny	ibm-granite/granite-4.0-h-small
Trace count	12	24
Layer count / hidden size	40 / 1536	40 / 4096
Decode positions	100, 500, 1000, 5000, 10000	100, 500, 1000, 5000, 10000, 20000
Bootstrap seed	20260508	20260508

Table 1: OutlierMigrate Phase 0 and Phase 1 gate results. Phase 0 values are from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json; strict decomposition values are from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/migration_decomposition.md. Phase 1 values are from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json; strict decomposition values are from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/migration_decomposition.md.

1e5c0b97e3c74fe7c1dd, and activation artifact SHA sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687; these are recorded in experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/model_provenance.json, experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json, and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json. The exact Phase 1 provenance values are: model snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39, prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e, and activation artifact SHA sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1; these are recorded in experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/model_provenance.json, experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json, and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json.

The Phase 0 checker returned PASS_OM_PHASE0_DECODE_TIME_MIGRATION. Its exact artifact reason was that the point estimate 0.81783854 is at least 0.05 and the CI lower bound 0.79726562 is greater than 0.05. The Phase 1 checker returned PASS_OM_PHASE1_REPLICATED_AT_SCALE. Its exact artifact reason was that the point estimate 0.84316565 is at least 0.05 and the CI lower bound 0.83343496 is greater than 0.05.

5.1 Phase 3: why simple migration-aware static protection is insufficient

Phase 3 tests the most direct intervention suggested by the measurement: instead of protecting only the position-100 top-1% channels, protect the union of top-1% channels observed

Quantity	Partial Phase 2 Nemotron-3
Decision	partial PASS; Qwen3.6/Kimi deferred
Strict set-leaving fraction	0.533713200380
Strict set-leaving 95% CI	[0.467414529915, 0.599982193732]
Within-set rank-shuffling fraction	0.269082383666
Within-set rank-shuffling 95% CI	[0.238841405508, 0.299056267806]
Original preregistered migration fraction	0.820809591642925
Original migration 95% CI	[0.7865325261158594, 0.8544931149097815]
Preregistered threshold	≥ 0.05 and CI lower > 0.05
Artifact complete	true
Model	nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16
Trace count	24
Layer count / hidden size	52 / 2688
Decode positions	100, 500, 1000, 5000, 10000, 20000

Table 2: Partial Phase 2 cross-family validation. The checker-backed packet is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z. This is not full cross-validation because Qwen3.6 and Kimi Linear are deferred.

Regime	Mean perplexity	Median perplexity	Median recovery	Recovery 95% CI
BF16 baseline	1.005358821083984	1.000282908378305	reference	reference
static-1% position 100	1.015228246226743	1.000282342709987	reference	reference
static-2% position 100	1.021467448964073	1.000390886047853	0.000000000000	[0.000000000000, 0.356017001423]
magnitude average top-1%	1.009248547916613	1.000236584155715	0.061276118929	[0.000000000000, 0.512245438462]
migration-aware union	1.008573512406127	1.000275313560728	0.000000000000	[0.000000000000, 0.711143244199]

Table 3: Phase 3 intervention and controls on Granite-4.0-H-Tiny. Values are from experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/per_trace_metrics.json, experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/metrics.json, and experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/control_metrics.json. Recovery is defined relative to the BF16-vs-static-1% gap.

at decode positions {100, 1000, 5000, 10000}. This is intentionally a static-set intervention. It asks whether membership migration alone explains enough of the quantization gap to support a simple deployment recipe.

The intervention kills. The checker returned the decision string KILL_OM_PHASE3_INTERVENTION_FAILS because the union regime had median recovery 0.000000000000, below the preregistered 0.20 kill floor. It also missed the pass rule: median recovery ≥ 0.50 with CI lower bound > 0.30 . The artifact packet is complete, and the control-stop condition did not fire. The best control, magnitude-average protection, exceeded union by 0.061276118929 median recovery, below the human stop threshold of 0.10. Ten of the 24 traces had no recoverable static gap under this scoring definition. That fraction, 0.416666666667, also exceeds the preregistered 25% no-gap kill condition. The checker reports the first kill reason it reaches, but the packet satisfies both kill paths.

The recovery statistic is intentionally centered on the per-trace BF16-vs-static-1% gap, so it can be negative when a regime is worse than static-1%, greater than one when a regime beats BF16 on the scored window, and zero when there is no recoverable static gap. In the union regime, 11 traces had positive recovery, 10 were zero because there was no recoverable static gap, 3 were negative, and 4 exceeded recovery 1.0. This unstable per-trace surface is part of the negative result rather than a hidden success case: the primary statistic was preregistered as the median across all 24 traces. The median was chosen for the gate because the recovery ratio can become very large when the static gap is small; a mean recovery would be dominated by unstable high-ratio traces rather than by the typical trace.

Grid	Positions	Median recovery	Recovery 95% CI
sparse	{100, 5000, 10000}	0.000000000000	[0.000000000000, 0.752281234025]
primary	{100, 1000, 5000, 10000}	0.000000000000	[0.000000000000, 0.711143244199]
dense	{100, 500, 1000, 2000, 5000, 7500, 10000}	0.000000000000	[0.000000000000, 0.920007799726]

Table 4: Position-grid sensitivity. Values are from experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/grid_sensitivity.md and experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/grid_sensitivity_metrics.json. Increasing grid density did not produce a nonzero median recovery.

Run / layer type	Layers	Strict set-leaving	Within-set shuffling
Phase 0 Granite-Tiny attention	4	0.618490	0.186198
Phase 0 Granite-Tiny SSM/Mamba	36	0.635995	0.174045
Phase 1 Granite-Small attention	4	0.557165	0.282266
Phase 1 Granite-Small SSM/Mamba	36	0.567243	0.269676
Phase 2 Nemotron-3 attention	6	0.563014	0.254630
Phase 2 Nemotron-3 MoE	23	0.526503	0.277778
Phase 2 Nemotron-3 SSM/Mamba	23	0.533280	0.264157

Figure 1: Layer-stratified migration decomposition from experimental/outlier_migrate/phase3/results/layer_stratified_migration.json. Values report layer-type means, averaged equally over layers. Strict set-leaving is the portion most directly relevant to static protected-channel membership; within-set shuffling motivates scale-refresh mechanisms but is not itself a membership failure.

5.2 Layer-stratified migration

The negative intervention result does not erase the measurement finding: it narrows the mechanism. Figure 1 uses the existing Phase 0, Phase 1, and partial Phase 2 activation packets to stratify strict set-leaving and within-set shuffling by layer type. The result is not confined to a single Granite mixer family. Granite shows strict set-leaving in both attention and SSM/Mamba layers, while Nemotron-3 shows it in attention, SSM/Mamba, and MoE-classified layers. This is consistent with the paper’s diagnostic claim: static membership is unstable in the measured packets, but successful handling likely needs more than a larger static protected set.

6 Discussion

On this Granite rank-migration surface, the Phase 0 result weakens a fixed position-100 outlier-map interpretation: under the preregistered metric, most position-100 top-1% channels do not remain rank-stationary through position 10000. Phase 1 promotes the dynamic-outlier hypothesis within the Granite family because the effect direction replicated on a larger model, twice as many traces, and an extended decode-position grid. This is not a first-observation claim about Mamba broadly. QMamba and OuroMamba already establish dynamic hidden-state or activation-outlier behavior in vision Mamba. The new evidence here is that the dynamic regime also appears under a frozen metric in hybrid LLM long-reasoning traces.

This is the strongest OutlierMigrate measurement evidence so far, but Phase 3 prevents a positive-method reading. The Nemotron-3 packet reduces the risk that the signal is merely a Granite-family artifact, but it remains partial: Qwen3.6 and Kimi Linear validation are deferred. More importantly, the predeclared migration-aware static protection recipe did not recover the BF16-vs-static-1% perplexity gap robustly. The measured migration is

therefore actionable as a diagnostic of why static maps may go stale, but not yet as a validated protection method.

The theoretical mechanism is plausible but not yet empirically validated outside Granite. In channel-wise gated recurrence or linear-attention layers, independent per-channel decay rates imply that channels retain state at different rates. Low-decay channels can accumulate magnitude over long decode, while high-decay channels shed magnitude. Under heterogeneous inputs and heterogeneous decay rates, the composition of the top 1% channels by magnitude should therefore change across decode positions rather than remain fixed to the position-100 set. This mechanism makes dynamic outlier migration a natural concern for static channel-protection schemes: a calibration-time set can be correct early in a trace and stale later in the same trace. Kimi Linear’s KDA, Qwen3.6-style Gated DeltaNet blocks, and GLA-style gated recurrence provide architectural examples where channel-wise or fine-grained data-dependent gating is a first-class operation (Moonshot AI Kimi Team, 2025; Qwen Team, 2026; Yang et al., 2024; 2023). This paragraph is a mechanism argument only; this draft has not measured migration on Kimi Linear, Qwen3.6, or GLA variants.

The Phase 3 negative result clarifies a bounded future-work direction. Simple set union handles only membership drift. It does not update quantization scales, does not react to within-set rank changes, and does not adapt layer-by-layer during decode. The layer-stratified analysis and the grid-sensitivity result suggest, but do not prove, that future methods may need online or decode-position-aware adaptation rather than a larger static protected set.

7 Limitations

This draft is not camera-ready final and should not be read as a positive systems method. It reports two passed observational measurement gates in one model family, one partial cross-family validation on Nemotron-3, and one preregistered negative intervention result. It still lacks Qwen3.6 and Kimi Linear validation, contamination audit, independent seed repeat beyond the recorded bootstrap procedure, and any intervention showing that migration-aware handling improves an application metric. The failed Phase 3 static-union intervention is now an explicit limitation rather than an absent experiment. The result may also depend on the chosen decode positions, top-channel fraction, and rank-delta threshold, all of which were fixed before analysis and should not be retuned on these packets.

The primary empirical measurement here is on Granite Mamba-2-style SSM hybrids. Nemotron-3 provides one additional hybrid family check, but it is not a complete substitute for validation on delta-rule linear-attention hybrids such as Qwen3.6 and Kimi Linear. Generalization to RWKV-7, GLA-style systems, and other efficient-recurrence variants is not established by these results. The paper also lacks a complete contamination or exploratory-history audit explaining the origin of the top-1% fraction, rank-delta threshold, decode positions, prompt slice, and model choices.

8 Reproducibility Statement

The Phase 0 result packet is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z`. It includes `environment.json`, `model_provenance.json`, `prompt_manifest.json`, `command_metadata.json`, `random_seed.json`, `metrics.json`, `bootstrap_ci.json`, `checker_result.json`, and `artifact_check.json`. The recorded Phase 0 command was:

```
experimental/shared/run_phase0_branch.py --branch outlier_migrate
```

The Phase 1 result packet is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z`. It includes the same artifact classes and records the Granite-4.0-H-Small snapshot commit `b8c0982bab7fde4eb48110f5a069527c008fab39`. The recorded Phase 1 command was:

```
experimental/outlier_migrate/phase1/run_om_phase1.py
```

The checker decisions and exact numbers in this draft are taken from the two Granite result directories above and the partial Phase 2 directory `experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z`. The OutlierMigrate runners emit the artifact-checked metric file `metrics.json`; they do not emit a separate `profiler_metrics.json`. The recorded Phase 2 command was:

```
experimental/outlier_migrate/phase2/run_cross_model.py \  
--partial-nemotron3-only \  
--run-id om_phase2_nemotron3_20260508T231723Z \  
--batch-size 4 --dtype bfloat16 --device auto
```

The Phase 3 intervention packet is `experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z`. It includes `environment.json`, `model_provenance.json`, `prompt_manifest.json`, `command_metadata.json`, `random_seed.json`, `per_trace_metrics.json`, `metrics.json`, `control_metrics.json`, `grid_sensitivity_metrics.json`, `checker_result.json`, and `artifact_check.json`. The recorded Phase 3 command was:

```
experimental/outlier_migrate/phase3/run_phase3_intervention.py \  
--run-id om_phase3_20260509T212000Z \  
--batch-size 4 --dtype bfloat16 --device auto \  
--reuse-prequant-run-dir \  
experimental/outlier_migrate/phase3/results/om_phase3_20260509T180200Z
```

The grid-sensitivity artifact was generated with:

```
experimental/outlier_migrate/phase3/run_grid_sensitivity.py \  
--run-dir \  
experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z
```

Figure 1 is a LaTeX-native summary table generated from `experimental/outlier_migrate/phase3/results/layer_stratified_migration.json`.

9 Ethics Statement

This is an activation-analysis measurement study on benchmark prompts. The current artifact does not make deployment, user-data, safety, systems-efficiency, or capability-improvement claims. The main ethical risk is overclaiming same-family Phase 0 and Phase 1 evidence, the partial Nemotron-3 packet, or the failed Phase 3 intervention as a validated cross-model positive method; this draft explicitly rejects that interpretation until deferred validation, audits, and a future intervention pass.

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